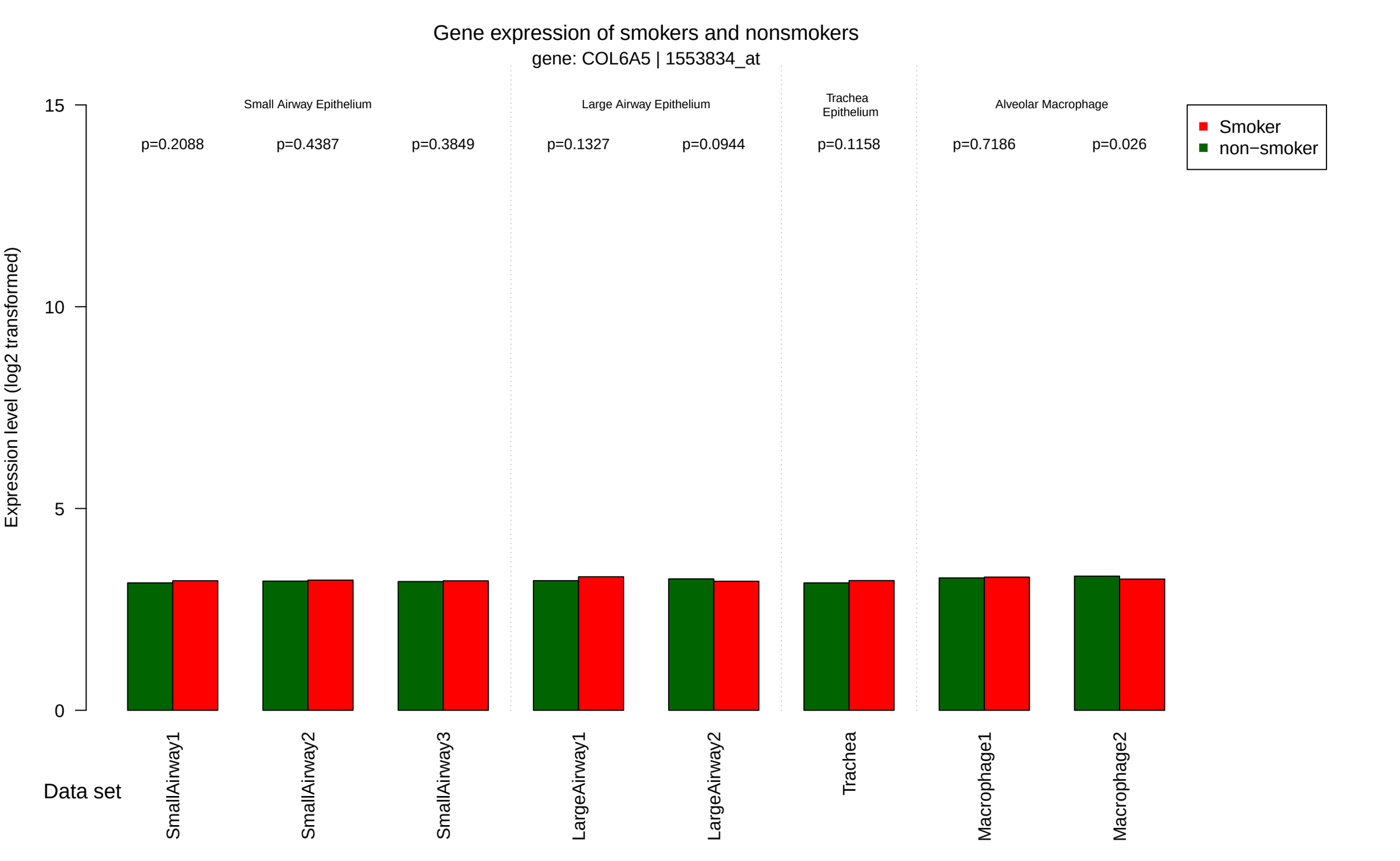
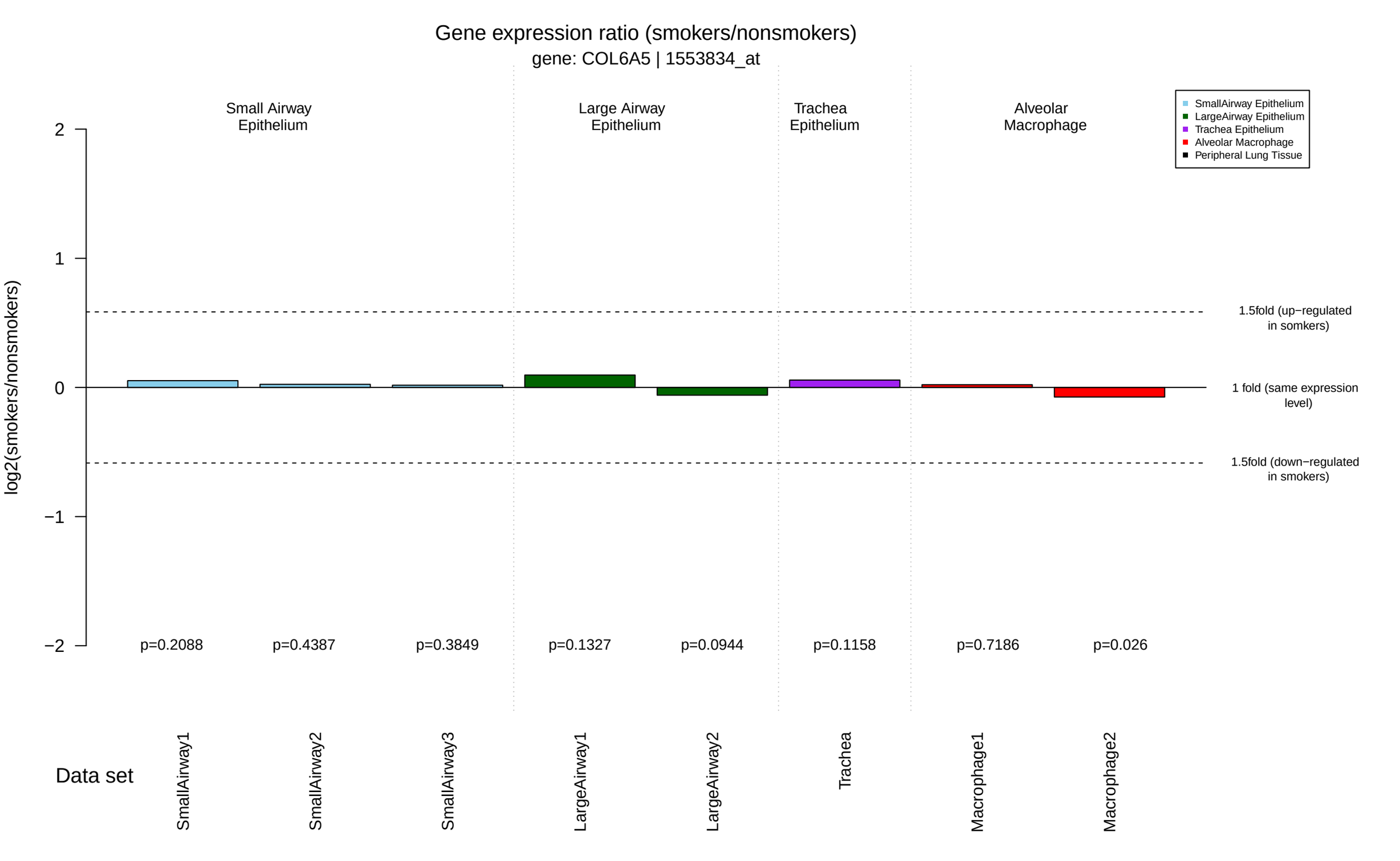
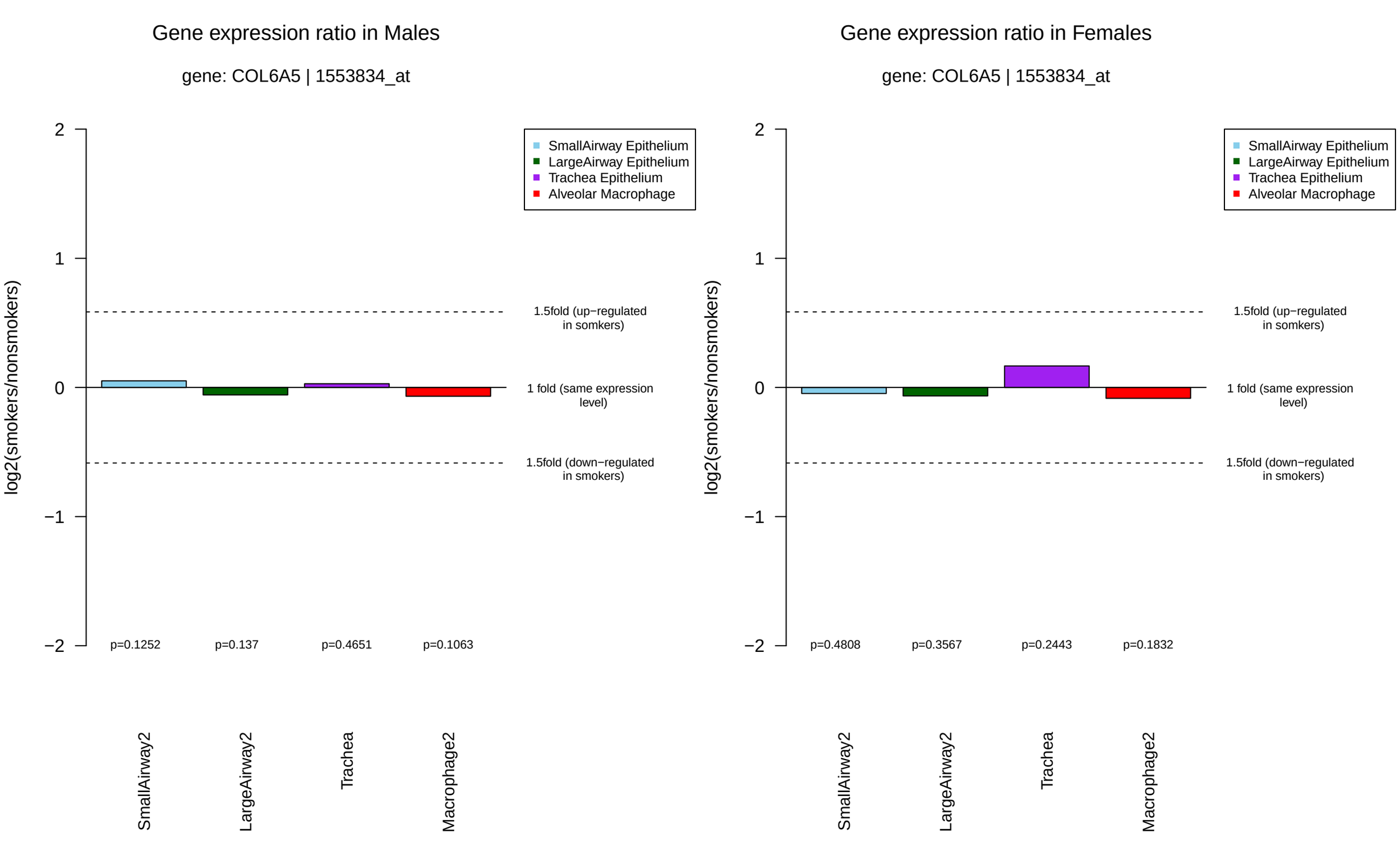
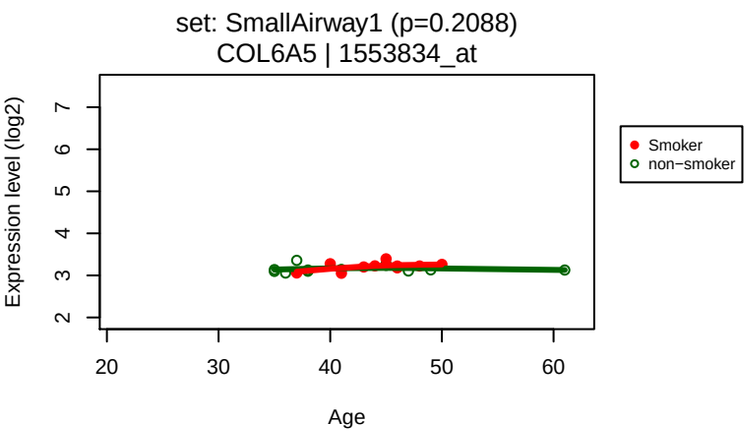
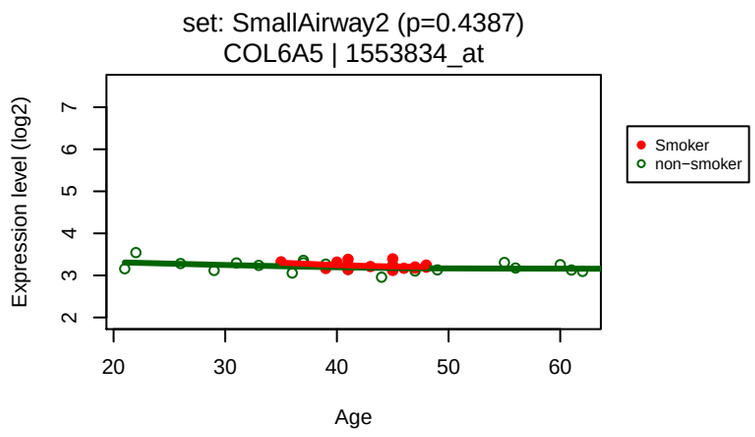
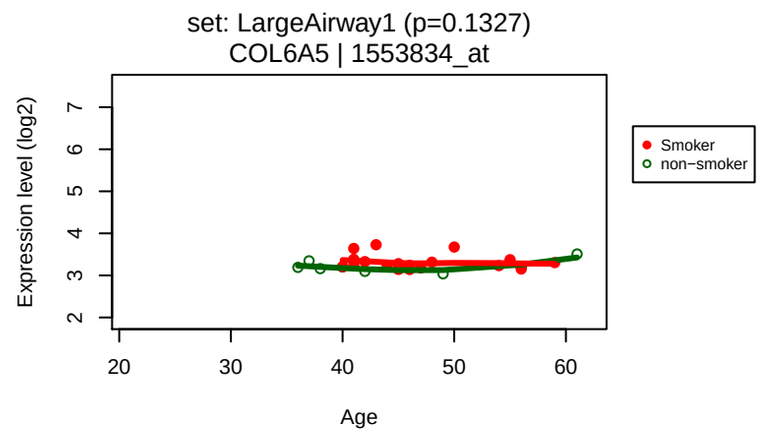

Gene expression of smokers and nonsmokers
gene: COL6A5 | 1553834_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2088
p=0.4387
p=0.3849
p=0.1327
p=0.0944
p=0.1158
p=0.7186
p=0.026
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

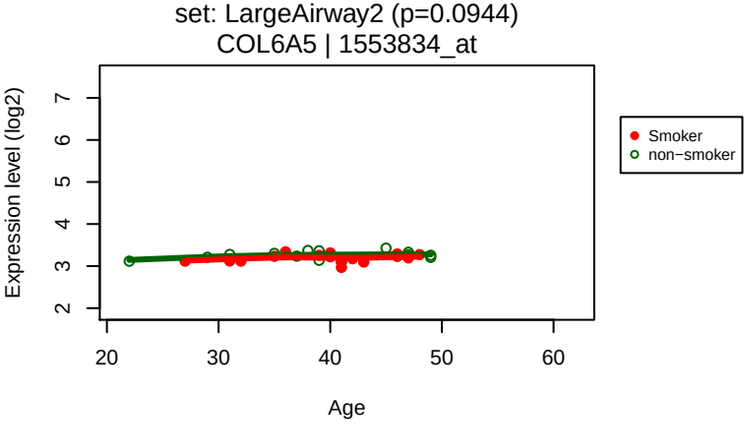

Gene expression ratio (smokers/nonsmokers)
gene: COL6A5 | 1553834_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2088
p=0.4387
p=0.3849
p=0.1327
p=0.0944
p=0.1158
p=0.7186
p=0.026
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

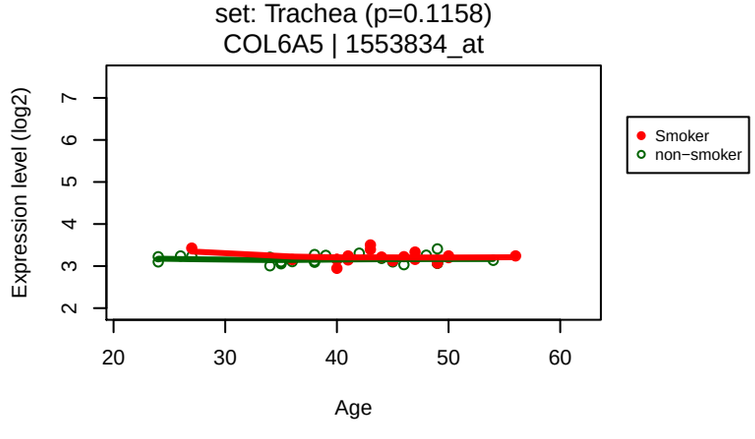

Gene expression ratio in Males
gene: COL6A5 | 1553834_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1252
p=0.137
p=0.4651
p=0.1063
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL6A5 | 1553834_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4808
p=0.3567
p=0.2443
p=0.1832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

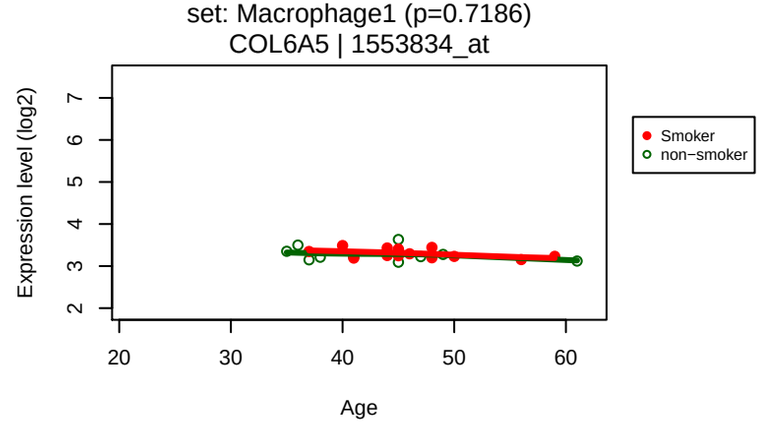

set: SmallAirway1 (p=0.2088)
COL6A5 | 1553834_at
Expression level (log2)
Age
Smoker
non−smoker

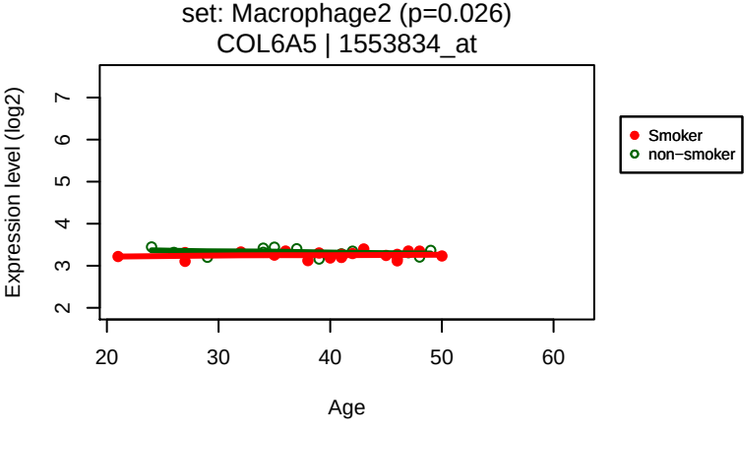

set: SmallAirway2 (p=0.4387)
COL6A5 | 1553834_at
Expression level (log2)
Age
Smoker
non−smoker

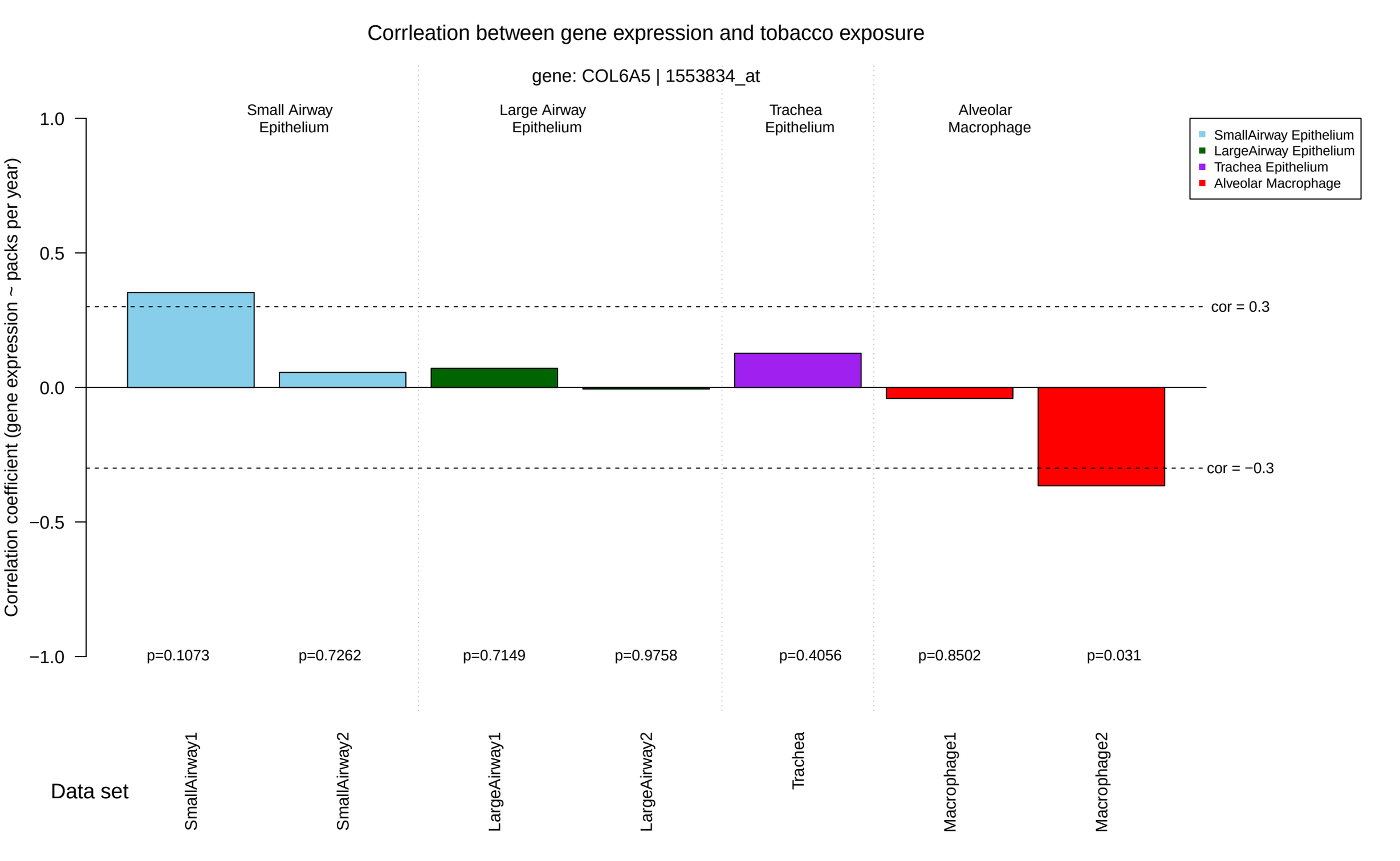

set: LargeAirway1 (p=0.1327)
COL6A5 | 1553834_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0944)
COL6A5 | 1553834_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1158)
COL6A5 | 1553834_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7186)
COL6A5 | 1553834_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.026)
COL6A5 | 1553834_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL6A5 | 1553834_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1073
p=0.7262
p=0.7149
p=0.9758
p=0.4056
p=0.8502
p=0.031
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2